Response to Reviewers — Second Round

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Selective Prediction and the Persistence Illusion: A Diagnostic Decomposition of VIX Regime Classification

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We thank both reviewers for their positive assessments and for the further suggestions, all of which we have implemented. No numerical results changed in this round; the edits are editorial additions and clarifications as detailed below.

Response to Reviewer 1

We are grateful for the reviewer's generous summary of the first revision and for the three remaining suggestions.

Comment 1 (Results sections could be shorter; numerical discussion duplicated by tables).

Response: We have trimmed the Results prose that restated table values. In particular, the main-results discussion (Section 5.1) and the persistence-quantification discussion (Section 5.4) now interpret the tables rather than repeat their entries, referring the reader to the tables for the numbers. We note that the larger structural version of this change was made in the first revision, when the threshold-sensitivity, VIX-threshold robustness, and sustained-label tables were moved to Appendix B; the present pass completes it at the paragraph level.

Comment 2 (captions should remind readers which accuracy is reported).

Response: We audited every figure and table caption and added a one-line clarifier wherever the base population was not already stated. Seven captions were amended (Figures 1, 3, and 6 and Tables 7 and 18–20) to state explicitly whether the reported quantity is selective accuracy on covered days, baseline accuracy on all test days, balanced accuracy, or an abstention rate over all days.

Comment 3 (when is selective prediction still beneficial from a risk-management perspective).

Response: We expanded the Practical Implications subsection with a dedicated risk-management passage describing three concrete deployment patterns that survive the paper's negative results: (i) *abstention as an uncertainty signal*—treating a spike in the abstention rate (which rises to 54% immediately before calm→high transitions at $N = 5$) as a trigger for hedge review, position-limit tightening, or de-risking, so that the system's silence rather than its directional calls carries the signal; (ii) *regime-stable confirmation*—using the 99–100% covered accuracy on stable days to gate strategies whose principal risk is false exit signals, such as systematic carry or volatility-selling programs; and (iii) *avoiding forced low-confidence calls*—routing abstained days to a human or a conservative default in automated pipelines. The passage closes with the deployment principle that selective systems should be used for what the diagnostic shows they can do (confirm continuation, flag their own uncertainty) and not for what they cannot (anticipate spikes).

Response to Reviewer 2

Comment (add more citations to strengthen the paper).

Response: We have added five references, woven into the Introduction and Related Work where they support the argument rather than appended as a list: Bekaert and Hoerova (2014, *Journal of Econometrics*) on the decomposition of the VIX into conditional variance and the variance risk premium; Bloom (2009, *Econometrica*) on uncertainty shocks as the macroeconomic driver of volatility spikes; Andersen, Bollerslev, Diebold and Labys (2003, *Econometrica*) on realized volatility modeling; Paye (2012, *Journal of Financial Economics*) on the limited out-of-sample value of macroeconomic predictors for volatility, which foreshadows our SHAP findings; and Gu, Kelly and Xiu (2020, *Review of Financial Studies*) as the benchmark for genuine machine-learning gains in empirical asset pricing, against which we position our message as complementary: flexible models can add value, but on highly persistent classification targets their apparent gains require persistence-matched auditing.

On the review-form item "Is the research design appropriate? — Must be improved."

Response: The reviewer's written comments raise no design concern, so we briefly address the checkbox here for completeness. The research design was substantially strengthened in the first revision in direct response to both reviewers and the editor: all baseline abstention thresholds are now calibrated on the training window only and frozen before test evaluation (eliminating the coverage-matching leakage identified in round one); the benchmark set was extended with XGBoost and a continuous-forecast-then-threshold HAR model; evaluation was made threshold-free via risk–coverage curves and AURC; inference robustness was documented across bootstrap block lengths of 32–90 days; and the complete pipeline was published at `https://github.com/akshatg10104/VIX_research` for independent verification. Reviewer 1, who reviewed these changes in detail, assessed the design as appropriate in this round. We would of course be glad to address any specific design concern the reviewer wishes to raise.

We thank both reviewers again for reviews that have materially improved the paper.

Sincerely,

Akshat Gupta and Jianguo Liu